



Privacy Notification

analyzing the user's emotions and reacting according to the real-time data without the user even having to actually do anything. This module provides intensity and evidence as the parameters which have an inverse relationship. While intensity provides data about the intensity of the perceived emotion, the evidence provides information in the form of a base 10 logarithmic scale on the probability of a particular expression being present.

The latest SDK has also been updated to enable compatibility with Nuance Dragon Assistant 3, which is supported on Windows 8 systems.

## BREAKDOWN OF THE PERCEPTUAL COMPUTING SDK

The Perceptual Computing SDK acts like an interface to help developers make the most of the integrated RealSense™ hardware. It provides pre-existing interfaces to algorithms, so that they can make the best use of these features and focus on developing the application rather than the algorithms for such interfaces, which usually

turn out to be very complex and time consuming, if written from scratch. The SDK supports a combination of various usage modes such as Speech Recognition, Facial Analysis, Emotion Sensing, Depth Tracking and Augmented Reality. Many of these features are possible with the help of the unique integrated camera specially built for capturing all this data. The SDK easily

allows features such as usage co-ordination, application co-ordination, privacy notification etc. It has been built in such a way that the loosely coupled interfaces provide room for integrating newer features without changing the structure too much – thus allowing for an extension of the usage modes in the future, since the software development kit is only expected to grow further during the course of time as we find new ways of capturing real-time data.

The SDK allows multiple applications to use the data at the same time, by defining one application as the master, which is allowed to control the hardware settings. All other applications using the hardware act as the slave for the master application. This is similar to applications using computing resources, i.e. the application with its window in focus will receive the highest priority while those running in the background can stop doing unimportant work.

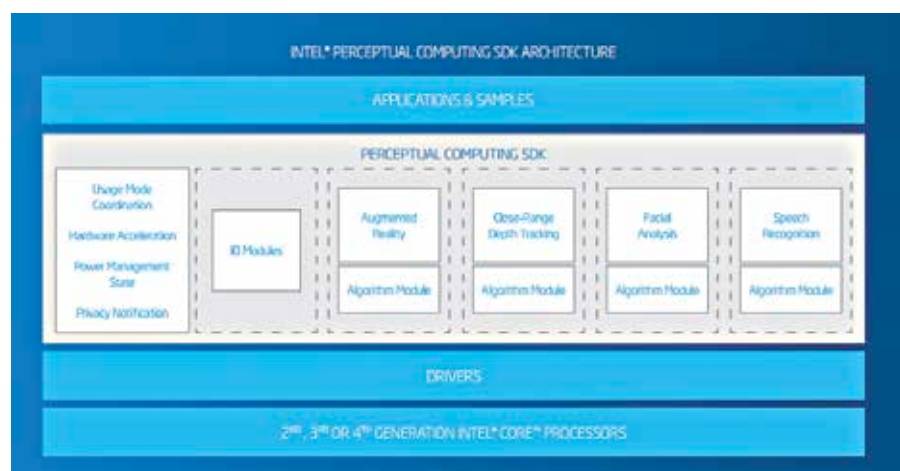
The privacy notification module performs the task of informing the end user whenever any potentially identifiable information is accessed and used. This information includes:

- Raw color or depth data from the camera
- Raw microphone data
- Face recognition data
- Face-based emotion detection data
- Face-based age detection data

Whenever a new user tries to use the application for the first time, a privacy notification dialog box is shown.

Apart from the above modules, the SDK has various algorithms available for usage:

- **Speech Recognition:** This allows for a specific application to understand pre-determined keywords spoken by the user.
- **Facial Analysis:** The face tracking module helps in determining the facial features and



Intel Perceptual Computing SDK Architecture